

Block Change Tracking

✓ **Why do we use BCT?**

Block Change Tracking (BCT) is used to speed up RMAN incremental backups.

Example

Suppose your database size is **2 TB**, but only **20 GB** of data changes every day.

Without BCT

- RMAN scans the entire **2 TB** database.
- Then it backs up only the changed 20 GB.
- Backup takes longer because of the full scan.

With BCT

- RMAN reads the BCT file to identify the changed blocks.
- It backs up only the changed **20 GB**.
- No full scan of the 2 TB database is needed.
- Backup completes much faster.

✓ **Advantages of BCT**

1. Speeds up RMAN incremental backups.
2. Reduces I/O on the database.
3. Reduces backup time.
4. Improves backup performance, especially for large databases.

✓ **How to enable BCT (Block Change Tracking)**

Step 1: Check if BCT is already enabled

SELECT status, filename FROM v\$dbblock_change_tracking;

```
STATUS
-----
DISABLED
```

Step 2: Enable Block Change Tracking

Option 1: Let Oracle create the BCT file automatically (recommended)

SHOW PARAMETER db_create_file_dest;

```
SQL> SHOW PARAMETER db_create_file_dest;
```

NAME	TYPE	VALUE
db_create_file_dest	string	

If it is not set, configure it:

ALTER SYSTEM SET db_create_file_dest='/u01/app/oracle/oradata' SCOPE=BOTH;

```
SQL> ALTER SYSTEM SET db_create_file_dest='/u01/app/oracle/oradata' SCOPE=BOTH;
System altered.
```

Then enable BCT:

ALTER DATABASE ENABLE BLOCK CHANGE TRACKING USING FILE
'/u01/app/oracle/oradata/bct/change_tracking.ctf' REUSE;

```
SQL> ALTER DATABASE ENABLE BLOCK CHANGE TRACKING
USING FILE '/u01/app/oracle/oradata/bct/change_tracking.ctf' REUSE; 2
Database altered.
```

The REUSE option allows Oracle to overwrite an existing BCT file with the same name.

Step 3: Verify

SELECT STATUS, FILENAME FROM V\$BLOCK_CHANGE_TRACKING;

```
SQL> SELECT STATUS, FILENAME FROM V$BLOCK_CHANGE_TRACKING;
```

STATUS	FILENAME
ENABLED	/u01/app/oracle/oradata/bct/change_tracking.ctf

Step 4: Disable BCT (if required)

ALTER DATABASE DISABLE BLOCK CHANGE TRACKING;

```
ALTER DATABASE DISABLE BLOCK CHANGE TRACKING;
```

Oracle removes the BCT file when it is disabled.